

Business Context

Bunnings makes up a large portion of Dulux's retail sales channel and is integrated with Dulux business systems via a process chain to provide weekly sales reports. This set of workflows ingest the flat files that Bunnings provides to Dulux into the different downstream systems.

Source and Target Systems

Solution Component	Description	Remarks
Bunnings SFTP server	Hosting server that contains the sales reports to be fetched by Dulux	This is the source systems and enforces 'Dual auth with basic and key exchange'.
SAP CI	Hosts Integration process that pulls files from bunnings SFTP server and put those in the SI file server via SFTP	
SI file server	This is the Dulux SFTP server that also acts as the file server for Dulux	All incoming files are staged here and copied across to downstream system including Local file archive, SAP Data Services server, SAP BW.
SAP Data Services	This process initialises a list of available files located in directory 'Receive\LatestInventoryFiles' and loads those into an Azure Data Lake.	This process exists due to an issue Dulux were facing connecting to the Azure storage account in question, and the Data Services tool was able to overcome that with a native driver included in the software package.
Data Engineering pipelines	This is a Azure Data Factory(ADF) that picks up the Bunnings file from the drop folder(\Bunnings_SalesOut) into Azure data lake. This pipeline process the raw file, and pushes the data to the Curated layer/in synapse views.	

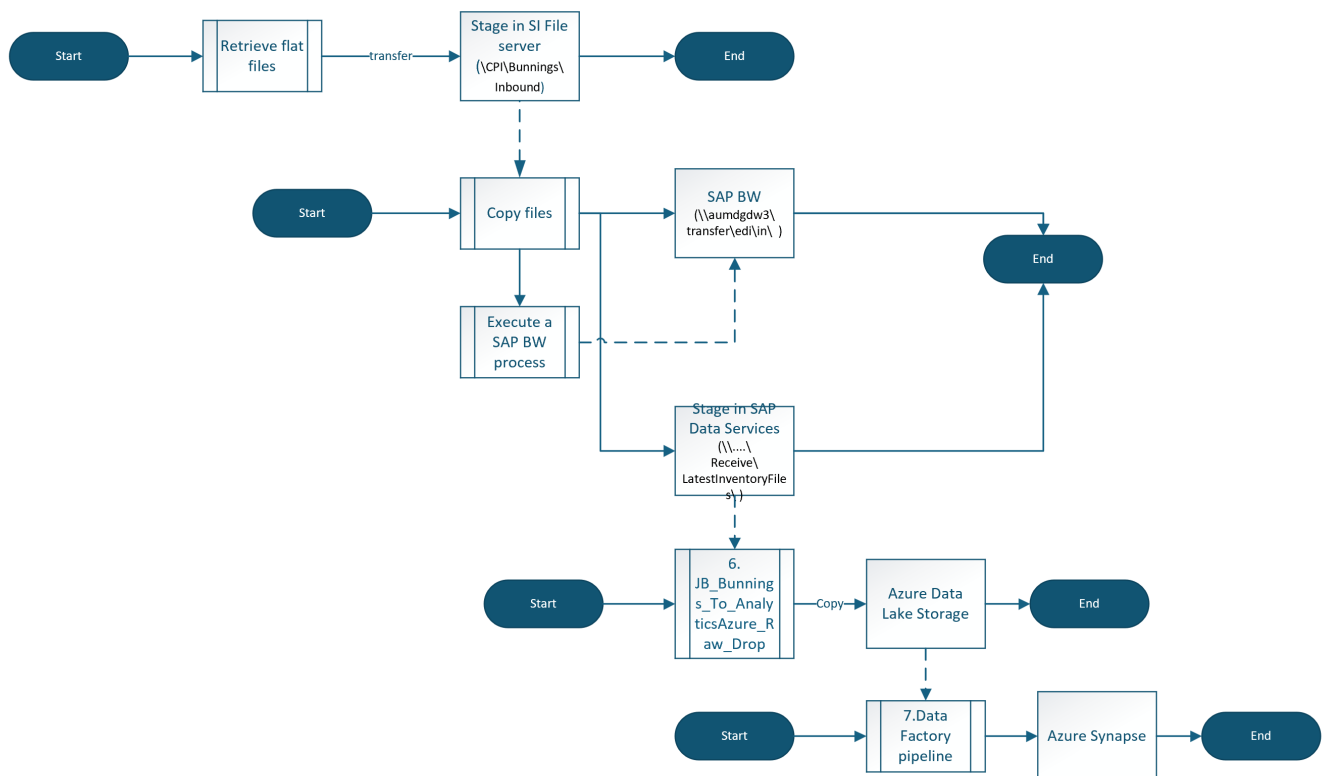
Assumptions

- Data provided as flat files from Bunnings, no filtering or transformations are required on the raw files.
- Only files that follow the filename pattern 'invrpt*' are fetched from the SFTP location.
- Files are received from the Bunnings at the same time(7:30pm) every day.

- The files are received in the same format each run and no extraneous files are received during pickup. Files are also received in Zip format.
- The ADF pipelines runs on a schedule and interacts with a azure sql database.
- No file needs to be saved as backup files in the SI file server(Path: 'F:\integrator\EDI\receive\backup\bunnings')
- The process running in the 'SAP Data Services platform' will be removed since Logic app will copy the required files from the staged location(i.e. azure storage account residing in the Integration Landing Zone) straight to the Azure data lake storage Gen2 storage account.
- The migration of this integration service to the new platform will be transparent to Bunnings.

Current State

Logical Design



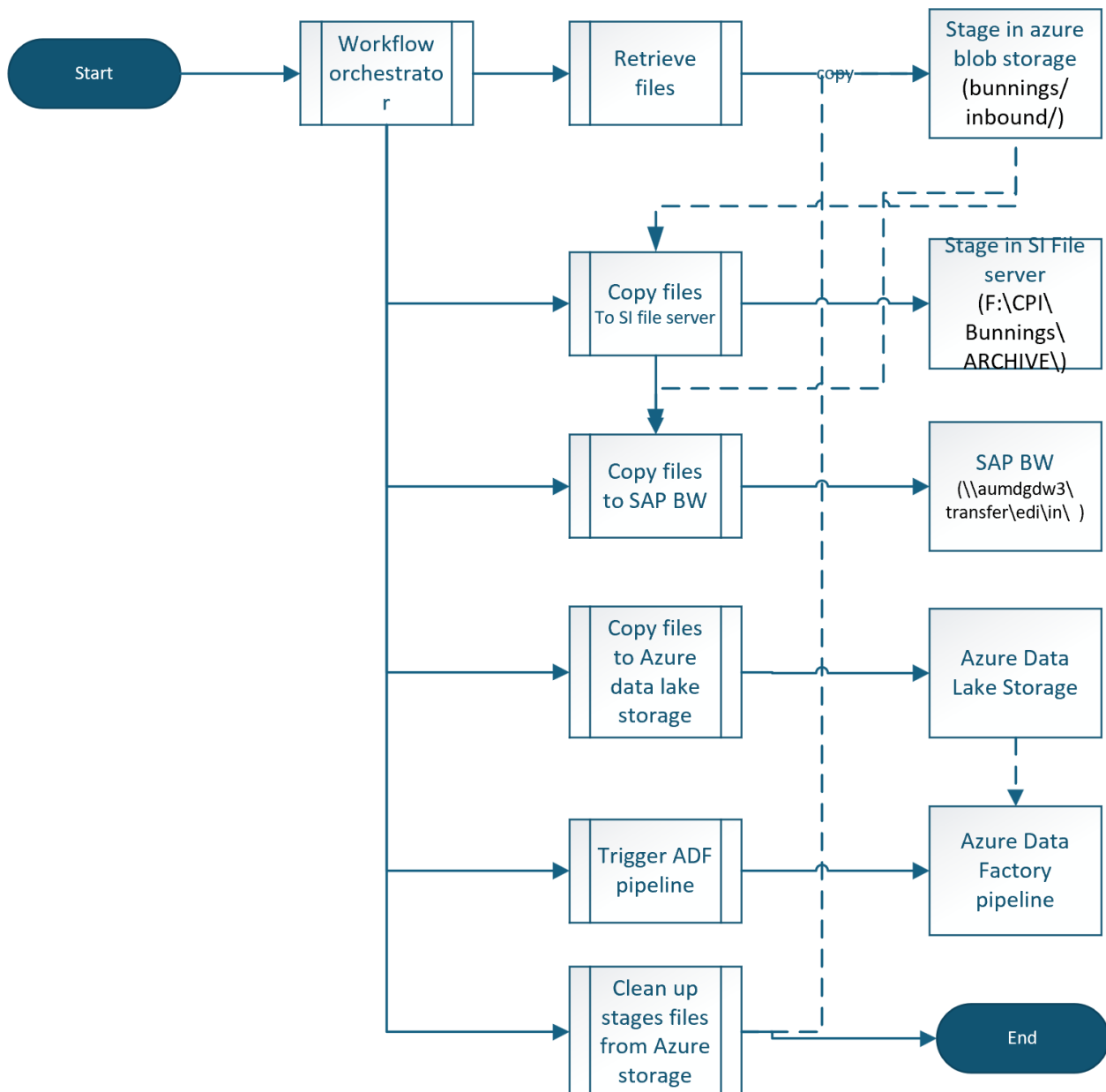
Key highlights:

- Dual auth with basic and key exchange is in use in the SFTP connectivity between bunnings SFTP server and SAP Integration flow.
- Basic auth is in use in the SFTP connectivity between SAP Integration flow and Dulux SFTP server(which is also the SI file server).
- The SI file server acts as the staging storage to facilitate file ingestion to the downstream business systems. It also retains the raw files as backup and archive.
- The overall process consists in multiple isolated sub-processes that perform file movements from the SI file server to different internal target systems.

- A powershell script runs in the SI file server that performs the file copy activity to the target business and data systems. It also triggers a job on the SAP BW system.

Target State

Logical Design



Key highlights:

- A timer triggered workflow will start the chain of execution and orchestrate the data movement process via nested workflows:
 - The 'Retrieve files' workflow will pull in the files from Dulux SFTP server and stage those in azure blob storage.
 - The 'Copy files to SI file server' workflow will copy all the staged files from blob storage and put the same in the SI File server as archive file (path: 'F:\CPI\Bunnings\ARCHIVE').

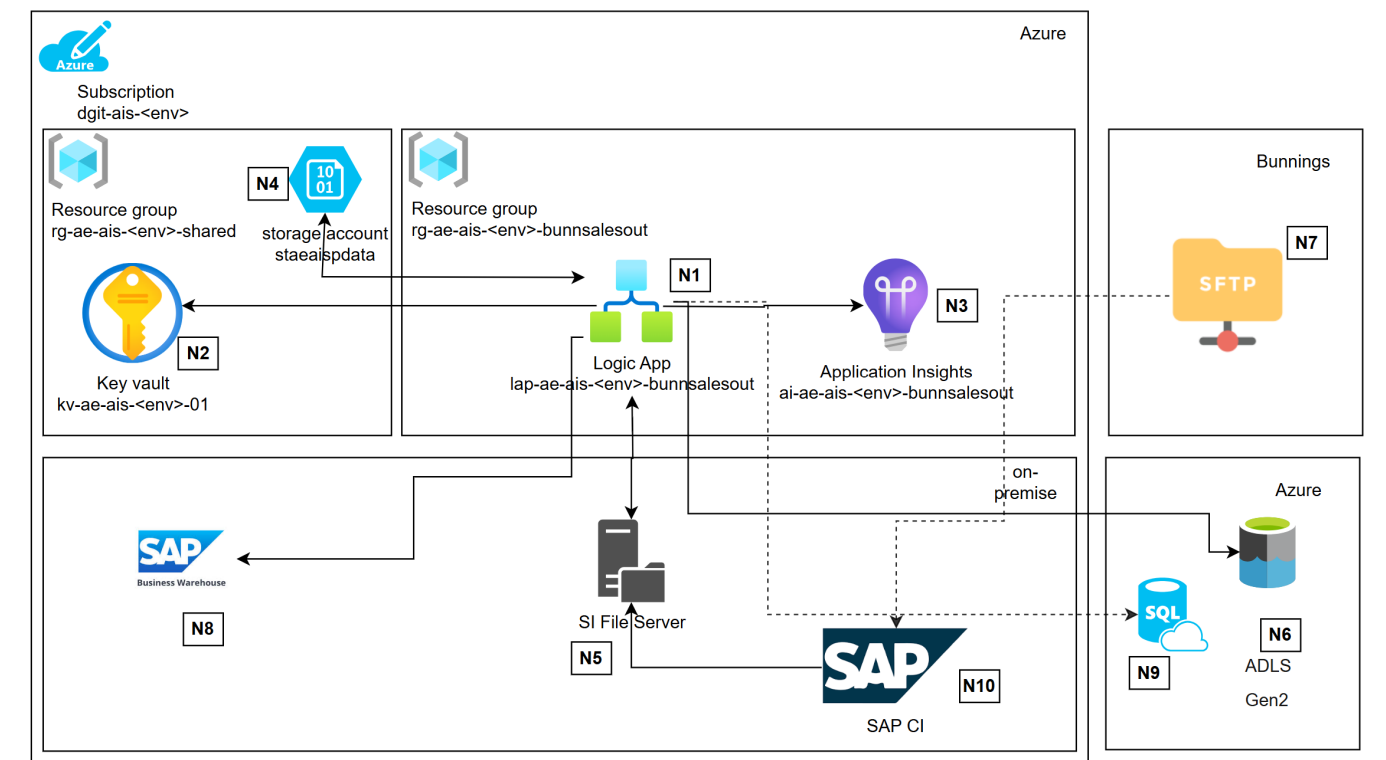
- 'Copy file to BW system' workflow will push the required set of files to SAP Business warehouse(BW) system and will trigger a SAP BW job via power shell script.
- 'Copy file to Azure Data Lake Storage' workflow will copy the files into Azure Data Lake storage.
- 'Trigger ADF pipeline' workflow will orchestrate the execution of the following two ADF pipelines:
 - It will trigger a data pipeline
 - will keep pulling the status of the job by doing a lookup in the control database.
 - Once the above job is complete , it will trigger the second data pipeline.
- Finally, a workflow will clean up the staged files from the blob storage.
- The logic app will connect with Azure Data Lake Storage Gen2 (ADLS Gen2) account by [Azure Blob Storage connector](#)
- Logic app will use 'File System built-in connector' to connect with the on-premises file shares(if required).
- Logic app will use office365 connector to send emails from b2b.support@duluxgroup.com.au.

Note:

- Some of the nested workflows can be run in a sequence or in parallel depending on the requirement. This will be discussed in detail during the implementation stage.
- We will use the Logic Apps PowerShell action to run a PowerShell script that will calls the batch file on the SI file server unless a more acceptable approach to 'call the SAP BW process' using a form of abstraction is found. The exact approach will need to be investigated before the build can commence.
- Recommendation is to maintain just one interface(and one protocol e.g. SFTP) with the SI file server (both to retrieve the files from bunnings as well as for the archiving).
- Any failure in relation to the file ingestion process to any downstream system(e.g. invalid file or error while uploading a file) will be reported to designated email address.

Physical Design

Resource Details

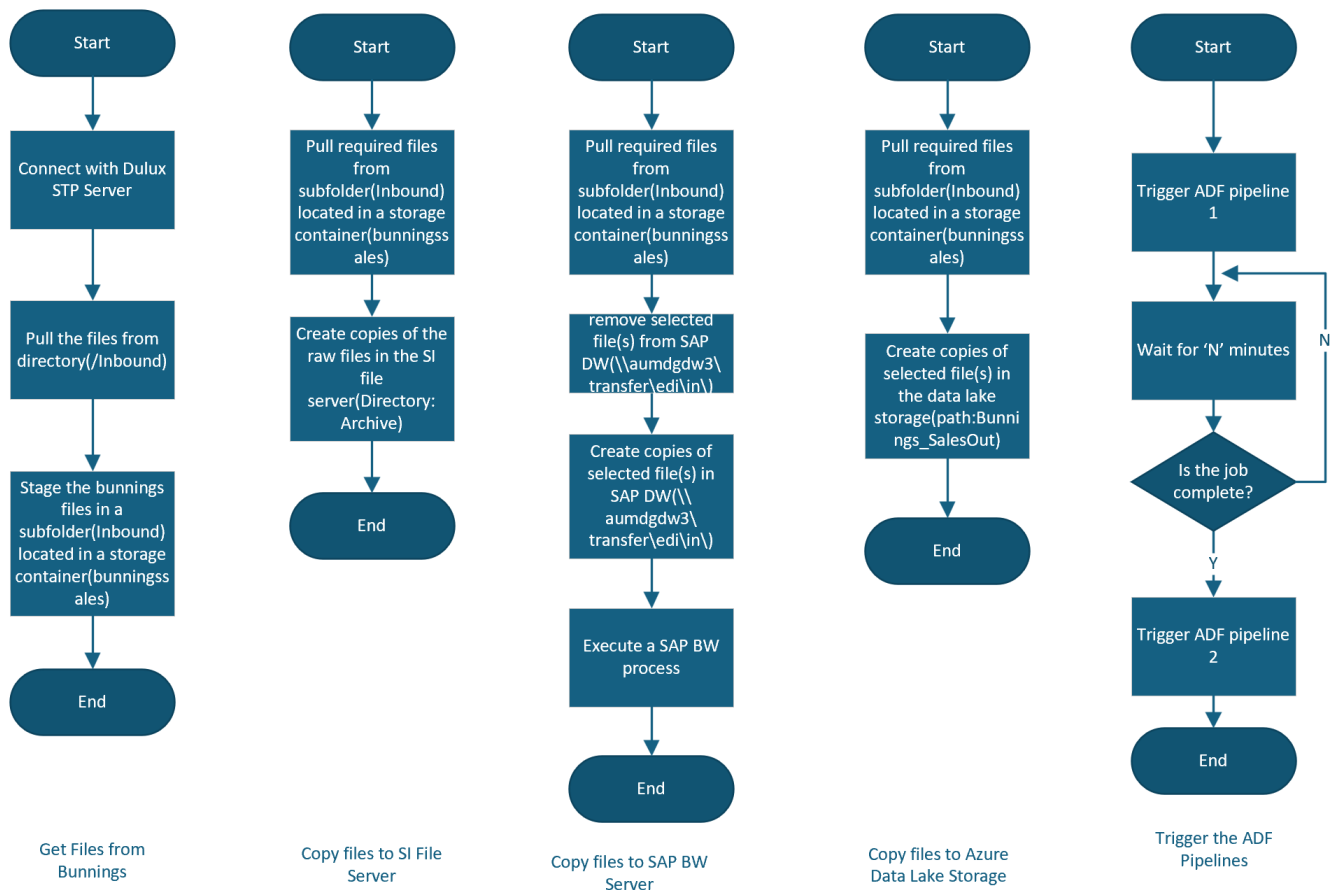


Reference	Component	Description
N1	Logic App	Logic app will be deployed in a new resource group. This will retrieve secrets (that are required for authorization with bunnings SFTP and SI file server) from the shared key vault (see N2). It will host different set of workflows to perform the 'copy file' activities to different target systems e.g. SI file server(see N5), SAP BW(see N8) and Azure data lake storage(see N6). The Logic app will use application insights resource (see N3) for logging and monitoring.
N2	Key Vault	Key vault shared by all integrations on the integration platform, expected to be used by logic app (see N1) for storing(and retrieving) secrets.
N3	Application Insights	Dedicated application insights resource used for monitoring of the logic app (see N1). Expected to be used for logs, metrics, and alerting.
N4	Azure blob Storage	It will be used as the staging storage where incoming files from Bunnings SFTP (see N7) will be fetched by the one logic app workflow (see N1)
N5	SI File Server	This acts as the persistent storage for Dulux where the raw files received from Bunnings SFTP are stored.
N6	Azure Data Lake Storage	Incoming files from bunnings are also pushed to this storage by another logic app workflow(see N1). These files are then picked up by data pipelines that pushes the data to the Curated layer/in synapse views.
N7	Bunnings SFTP Server	This is the source system that publishes the sale reports in an SFTP location path. This server is operated by Bunnings and only vetted systems are granted access to it.

Reference	Component	Description
N8	SAP BW	SAP Business Warehouse system where sales files from bunnings will need to be dropped followed by execution of a SAP BW process.
N9	Control Database	This is an Azure SQL database that contains the processing status of different jobs that are managed by data pipelines.
N10	SAP CI	Hosts the integration process that pulls files from Bunnings SFTP server and puts them in the SI file server via SFTP.

Integration Workflows

Following diagram illustrates the set of actions that will be orchestrated by the parent logic app workflow:



Access and permission Requirement

- Access to the mailbox: b2b.support@duluxgroup.com.au.
 - A service account will need to be created and a compatible office365 license needs to be assigned.
- The managed identity of the Logic app needs to be granted required 'Role Based Access Control'(RBAC) permission to the azure data lake storage account.

- The managed identity of the Logic app needs to be granted required 'Role Based Access Control'(RBAC) permission to execute a pipeline in the existing ADF.

Connectivity Requirement

Connectivity Requirement for DEVELOPMENT Environment

Source System	Source IP	Destination System	Destination IP	Destination Port	remarks
Logic App	<ase subnet>	Bunnings SFTP server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Dulux SI File server	<To be confirmed >	22,445 <To be confirmed by Dulux>	
Logic App	<ase subnet>	SAP BW server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Azure Data Lake Storage Account	<To be confirmed >	443	
Logic App	<ase subnet>	Azure Sql Database	<To be confirmed >	1433	

Connectivity Requirement for TEST Environment

Source System	Source IP	Destination System	Destination IP	Destination Port	remarks
Logic App	<ase subnet>	Bunnings SFTP server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Dulux SI File server	<To be confirmed >	22,445 <To be confirmed by Dulux>	
Logic App	<ase subnet>	SAP BW server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Azure Data Lake Storage Account	<To be confirmed >	443	
Logic App	<ase subnet>	Azure Sql Database	<To be confirmed >	1433	

Connectivity Requirement for Production Environment

Source System	Source IP	Destination System	Destination IP	Destination Port	remarks
Logic App	<ase subnet>	Bunnings SFTP server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Dulux SI File server	<To be confirmed >	22,445 <To be confirmed by Dulux>	
Logic App	<ase subnet>	SAP BW server	<To be confirmed >	<To be confirmed by Dulux>	
Logic App	<ase subnet>	Azure Data Lake Storage Account	<To be confirmed >	443	
Logic App	<ase subnet>	Azure Sql Database	<To be confirmed >	1433	

Security

- Only required access will be granted to the logic app over the azure data lake storage account, Azure data factory and the file server(to execute the SAP BW process).
- The bunnings sales files will be removed once the data ingestion is complete. Or, there can be a logic app workflow that will clean up the files after 'N' number of days from the date these were created.

Identity

- Managed identity will be used to access azure resources e.g. Azure Data Lake Storage and ADF pipeline.

References

- [Connect to on-premises file systems from workflows in Azure Logic Apps](#)